



الاختبار النصفى لمقرر التعلم العميق المتقدم (AI4204)

Midterm Exam – Advance Deep Learning (AI4203)

الفصل الدراسي	تاريخ الاختبار	مدة الاختبار
472	الأربعاء 1447/10/20 الموافق 2026/04/08	ساعة

تعليمات الاختبار:

- عدد أسئلة الاختبار ١٢ سؤالاً.
- درجة الاختبار ٣٠ درجة.
- تأكد من كتابة اسمك ورقمك الجامعي ورقم الشعبة في ورقة الإجابة.
- غير مسموح باستخدام أي مذكرات أو مراجع أثناء الاختبار (Closed book exam).
- في حال احتجت إلى مساعدة أو استفسار، ارفع يدك وانتظر المراقب للرد علي ك.
- احرص على متابعة وقت الاختبار لكي تتمكن من الإجابة على جميع الأسئلة

Exam Instructions

- The exam consists of 12 questions.
- The total exam score is 30 marks.
- Make sure to clearly write your name, university ID number, and section number on the answer sheet.
- No notes or reference materials are allowed during the exam (closed-book exam).
- At the end of the exam, submit both the question paper and the answer sheet to the invigilator.
- If you need assistance or have a question, raise your hand and wait for the invigilator to assist you.
- Please manage your time carefully to ensure you are able to answer all required questions.

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1. What are the three main parts that make up the Transformer architecture?

Well we have ~~three~~: Embedding and Positional Encoder and Attention (includes FC)
 its also consist of: (Embedding and Pos and encoder and decoder)

2. What is the primary function of "Positional Encoding" in Transformers?

To Add the Position of current token or word embed

Because the Transformer goes in parallel Not sequential it has No sense of order so we add Position encoding Because the order effect the meaning and grammar

3. In calculating self-attention, three variables taken from database terminology are used. What are they?

- Query: What's the word looking for from other
- Key: What's the word here and can give
- Value: What's the ~~meaning~~ value of a word

4. In the self-attention mechanism, what does the Query (Q) represent?

- A) The stored information retrieved as a result of the attention process x
- B) The current word that is "asking" which other words are most relevant to it
- C) A label used to index and match words in the attention lookup table
- D) The scaling factor applied before the SoftMax to stabilize gradients

$$\text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

5. The SoftMax function in the self-attention equation converts similarity scores into probabilities that sum to 1, producing attention weights for each word.

A) True

B) False



6. What is the key difference between Self-Attention and Masked Self-Attention regarding access to other words?

- A) Self-Attention can only look at previous words, while Masked Self-Attention can look at all words
- B) Self-Attention allows every word to attend to all other words, while Masked Self-Attention restricts each word to attend only to previous words
- C) Both allow full access to all words, but Masked Self-Attention applies dropout for regularization
- D) Masked Self-Attention attends to future words only, while Self-Attention attends to past words only

7. Why do we transpose the Key matrix (K) when multiplying it with the Query

- A) To normalize the values and prevent large numbers
- B) To make the matrix dimensions compatible for multiplication
- C) To reduce the size of the matrix for faster computation
- D) To apply positional encoding to the input

8. What is the title of the seminal 2017 paper that first introduced transformer architecture, and what specific task was the model originally designed to perform?

"Attention is All you need"
it was for machine translation between English → Spanish or French

9. In Encoder-Decoder Attention, where do Q, K, V come from?

Query: comes from the decoder

Key, Value: come from the encoder

10. Mention two practical uses for the "Context Aware Embeddings" produced by Encoder-Only Transformers?

- Sentiment Analysis: for example happy or sad sent

- Named entity Recognition for example Noun - ~~the~~



3.5

11. Explain what Word Embedding and Positional Encoding are, and how they are combined to form the input of a Transformer model.

Word Embedding: it's a trained Neural Network that assign Each token to a semantic vector for example in that vector "King - man + Women = Queen" But in transformer it has No sense of Position.

Position Encoding: Because word embedding has No sense of order and because order do matter (we add $\sin$ and $\cos$ waves to the word Embedding representing the order)

12. Compare Encoder-Only and Decoder-Only Transformers. Explain the type of attention used in each, their primary goals, and why ChatGPT is considered a Generative Model.

Aspect	Encoder-Only	Decoder-Only
Attention	Self Attention	masked Attention
Goal	understand the text and produce text context aware	Next token Prediction
Example tasks	Sentiment analyse OR BERT model 2. احسنه	Gemini or GPT for writing a story Task - احسنه

Why ChatGPT is generative:

Because it's a Decoder only model that generate token By token and only depend on current and previous token

~ Good Luck ~



Questions Overview (Marks Distribution)

Q#	Question Type	Marks
Q1	Short Answer	2
Q2	Short Answer	2
Q3	Short Answer	2
Q4	MCQ	2
Q5	True/False	2
Q6	MCQ	2
Q7	MCQ	2
Q8	Short Answer	3
Q9	Short Answer	2
Q10	Short Answer	2
Q11	Essay	4
Q12	Essay	5
Total		30